

Market Crash Alert System Using Machine Learning

Final Project Report

Abstract

Financial market crashes can result in significant economic losses for investors and institutions. Predicting such events is difficult because financial markets are influenced by numerous interconnected factors, including market volatility, commodity prices, macroeconomic indicators, and investor sentiment. This project presents a Market Crash Alert System that estimates the probability of a future crash in the Nifty 50 index over the next 21 trading days.

The system uses four ensemble machine learning algorithms—CatBoost, LightGBM, XGBoost, and Random Forest—whose outputs are calibrated using Platt Scaling and combined into a single blended probability. The application predicts the likelihood of three crash severities:

- Greater than 5%
- Greater than 8%
- Greater than 10%

A Streamlit web application was developed to present these predictions in an interactive dashboard with market statistics, probability trends, and risk indicators. The project demonstrates how machine learning can be applied to financial risk forecasting rather than price prediction.

1. Introduction

Stock market crashes are rare but highly impactful events that can erase years of investment returns within days. Traditional forecasting methods often struggle because crashes are nonlinear and influenced by multiple interacting variables.

Instead of forecasting the exact future index value, this project predicts the probability that the Nifty 50 will experience a significant drawdown within the next 21 trading days.

The project integrates:

- Historical market data
- Macroeconomic indicators
- Technical indicators
- Ensemble machine learning

- Probability calibration
- Interactive visualization

into one complete decision-support system.

2. Objectives

The primary objectives of this project are:

- Predict future market crash probability instead of future prices.
- Develop separate models for multiple crash severities.
- Improve prediction reliability using an ensemble of machine learning models.
- Convert raw model outputs into calibrated probabilities.
- Build an interactive Streamlit dashboard for investors.
- Provide interpretable risk levels for decision making.

3. Dataset

The project uses publicly available financial data downloaded through Yahoo Finance.

Primary Market

- Nifty 50 Index (^NSEI)

Macroeconomic Variables

- India VIX
- Gold Futures
- Crude Oil Futures
- USD/INR Exchange Rate

Historical data beginning from 2010 is used for training.

4. Problem Formulation

Instead of regression, the problem is formulated as a binary classification task.

For every trading day, the future minimum price over the next 21 trading days is calculated.

If the future decline exceeds a threshold:

- 5%
- 8%
- 10%

then the observation is labeled as: **Crash = 1**. Otherwise, **Crash = 0**.

Thus three independent datasets are created for three different crash definitions.

5. Feature Engineering

Rather than using raw prices directly, several engineered features were created.

Market Returns

- 1-day return
- 5-day return
- 21-day return
- 63-day return

Volatility

- 21-day rolling volatility
- 63-day rolling volatility

Drawdown Features

- 21-day drawdown
- 63-day drawdown
- 252-day drawdown

Trend Indicators

- EMA 50
- EMA 200
- Distance from EMA50
- Distance from EMA200

Volatility Indicator

- Average True Range (ATR)

Macroeconomic Features

- 5-day VIX return
- 21-day VIX return
- Gold return
- Crude Oil return
- USD/INR return

These engineered variables provide information about trend, volatility, momentum and external market conditions.

6. Machine Learning Models

Four different ensemble learning algorithms were trained.

CatBoost

Gradient boosting algorithm designed for robust performance on structured datasets.

- Handles nonlinear relationships
- Resistant to overfitting
- Performs well with limited feature preprocessing

LightGBM

Histogram-based gradient boosting framework.

- Fast training
- Efficient memory usage
- Excellent performance on tabular financial data

XGBoost

Extreme Gradient Boosting algorithm.

- Strong predictive performance
- Regularization reduces overfitting
- Handles class imbalance effectively

Random Forest

Bagging-based ensemble algorithm.

- Robust baseline model
- Less sensitive to noisy observations
- Good generalization ability

7. Hyperparameter Optimization

Instead of manually selecting parameters, every model was optimized using Optuna.

Optimization settings:

- 100 optimization trials
- TimeSeriesSplit cross-validation
- 21-day gap to prevent data leakage
- ROC-AUC used as optimization objective

Each crash threshold (5%, 8%, and 10%) has its own optimized hyperparameters.

8. Time Series Cross Validation

Since stock market data is chronological, ordinary random train-test splitting would leak future information.

The project therefore uses **TimeSeriesSplit** with:

- Chronological training
- Chronological testing

- 21-day gap between train and validation

This makes evaluation more realistic.

9. Probability Calibration

Financial crash datasets are highly imbalanced because crashes are relatively rare.

To compensate for imbalance, the models use:

- Balanced class weights
- Scale positive weight

Although these improve classification accuracy, they distort probability estimates.

Therefore every model is calibrated using **Platt Scaling**, implemented through:

```
CalibratedClassifierCV(method="sigmoid")
```

This converts raw model scores into well-calibrated probabilities.

10. Ensemble Learning

Each calibrated model predicts the crash probability independently.

Final crash probability is calculated as the average of:

- CatBoost probability
- LightGBM probability
- XGBoost probability
- Random Forest probability

Blending multiple models reduces prediction variance and improves robustness.

11. Streamlit Application

An interactive web application was developed for end users.

Features

- Live market data retrieval
- Daily model retraining using the latest data
- Predictions for 5%, 8%, and 10% crashes
- Blended crash probability
- Individual model probabilities
- Risk categorization
- Interactive probability charts
- Market snapshot
- Historical prediction table

The dashboard is designed to provide an intuitive overview of current market conditions and forecasted crash risks.

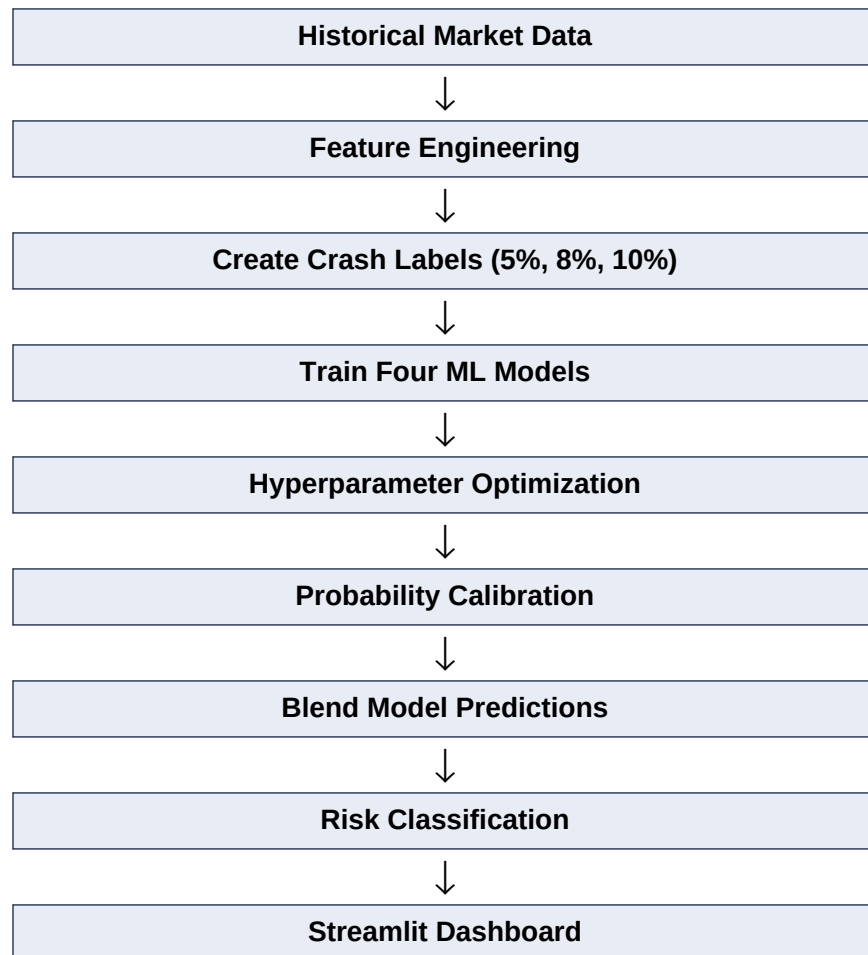
12. Risk Categories

Predicted probabilities are mapped into four intuitive risk levels:

Probability	Risk Level
< 15%	Low Risk
15 – 35%	Moderate Risk
35 – 60%	High Risk
> 60%	Extreme Risk

These categories allow users to interpret model predictions without needing statistical expertise.

13. Project Workflow



14. Applications

This system can be used for:

- Investment risk monitoring
- Portfolio management
- Asset allocation
- Hedging decisions
- Early warning systems
- Financial research
- Educational demonstrations of ML in finance

It is intended as a decision-support tool, not an automated trading system.

15. Advantages

- Uses multiple machine learning models
- Employs probability calibration for more reliable outputs
- Incorporates macroeconomic and technical indicators
- Time-series aware validation reduces leakage
- Interactive visualization through Streamlit
- Predicts probabilities rather than binary outcomes
- Daily retraining with updated market data

16. Limitations

- Uses only historical market information; unexpected geopolitical or economic shocks cannot be anticipated.
- Yahoo Finance data may contain missing values or revisions.
- Market regimes evolve over time, requiring periodic retraining.
- Crash events are inherently rare, leading to class imbalance.
- Predictions represent probabilities rather than certainty.

17. Future Enhancements

Possible improvements include:

- Incorporating additional macroeconomic variables such as interest rates and inflation.
- Adding sentiment analysis from financial news and social media.
- Using explainable AI techniques such as SHAP values.
- Experimenting with stacking or weighted ensemble methods.
- Supporting international stock indices.

- Providing email or mobile push notifications when crash probabilities exceed user-defined thresholds.
- Backtesting portfolio strategies driven by the generated risk signals.

18. Conclusion

This project presents a machine learning–based Market Crash Alert System for forecasting the probability of significant declines in the Nifty 50 index over the next 21 trading days. By combining CatBoost, LightGBM, XGBoost, and Random Forest models, optimizing them with Optuna, calibrating their probabilities using Platt Scaling, and averaging their outputs, the system produces robust and interpretable crash-risk estimates. The accompanying Streamlit application delivers these forecasts through an interactive dashboard that includes live market data, probability trends, and risk categories, making the results accessible to both technical and non-technical users. While the system cannot predict every market event, it provides a practical early-warning framework that can support investment decisions and financial risk management.